

The Stark-Zeeman Line-Shape Code

User Manual and Theoretical Guide

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Open-Source Coupled Stark-Zeeman Plasma Line-Shape Project

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Numerical and physical contract after the follow-up audit

Stored profiles are densities per eV, proportional to squared dipole strength with equal unnormalized substate weights. They are not absolute emissivities: state populations, photon-energy factors and the radiometric prefactor are not supplied. Angular frequencies in this derivation must be multiplied by $\hbar/e$ before comparison with code widths, fluctuation energies or bin tolerances in eV. Numerical angular momentum matrices use dimensionless quantum numbers; radial integrals are numbers in Bohr-radius units. The SI quadratic-Zeeman energy is divided by the elementary charge to obtain eV, applying a_0^2 only once.

Empirical physical broadening and result detunings use the fixed zero-field shell-trace reference, not the field-dependent eigensolver centering shift. Tabulated isotope energies do not make a multi-electron atom hydrogenic; only complete supported hydrogen-like data can be used. Natural damping is state resolved: E1 rates are summed over all lower shells, final substates and photon polarizations in the uncoupled basis, then the diagonal rate operator is rotated into each Stark-Zeeman eigenbasis. Transition $i \rightarrow j$ receives $\hbar(\Gamma_{u,i} + \Gamma_{l,j})/2$. The compatibility option `natural_width_mode='shell_average'` restores the historical common width. These rates use gross-structure transition energies and neglect their fine-structure corrections. Comparison with the critically evaluated H I multiplet probabilities of Wiese and Fuhr [17] gives agreement within 0.1% for the 2p, 3s, 3p and 3d total E1 rates; the machine-readable inputs are in `starkzee/data/radiative_benchmarks.json`. These are E1-only rates: the zero returned for 2s excludes its finite two-photon and other non-E1 decay channels and is not a prediction of infinite physical lifetime.

Constant electron widths with Doppler are evaluated with direct Voigt kernels, including transition-dependent diagonal-operator widths. Frequency-dependent widths are never silently replaced by their resonance value; the legacy force flag is a no-op. For FFT Doppler, nonuniform monotonic input is evaluated on a uniform increasing energy grid no coarser than its smallest spacing, then interpolated back. Refine both grid and window: sampling unresolved narrow components or cropping tails can lose area. Gaussian helper kernels use zero lag at index $N/2$ rounded down, not the physical line center, and zero extension. Coordinate conversion of spectral density requires the absolute Jacobian; the high-level spectral-density accessor supplies vacuum-nm, THz and inverse-cm densities without changing the stored per-eV arrays.

Field weights are $W_i = P(\beta_i)\Delta\beta / \sum_j P(\beta_j)\Delta\beta$. Converge field cutoff, field resolution and angular resolution independently. Potekhin neutral and charged formulas are different

fits, not interchangeable; the displayed screened formula below is the charged branch, and the unscreened rational CDF is the neutral branch. The code uses classical electron Debye screening for Potekhin, not a general degenerate or multitemperature screening model.

Hamiltonian diagonalization is exact only within the retained single shell. No universal magnetic-field or density range guarantees accuracy: compare inter-shell couplings to shell gaps, relativistic corrections to retained terms, collision couplings to dressed gaps, and fluctuation energies to component widths. Low density does not imply Doppler dominance without temperature and lifetime information; high density alone does not establish quasi-static breakdown. Compact-object spectra are not validated by this truncation.

Full versus projected dipole-square closures, lower-manifold and interference terms, and the collision-frequency cutoff convention require matched primary-code benchmarks. Suppression of the weak-collision G factor alone does not bound the strong-collision contribution. PPPB's printed cutoff uses $2\pi/\tau_e$, whereas this implementation retains its historical $1/\tau_e$ convention; exact PPPB reproduction is not claimed. The stated strong-collision constants for $n \geq 2$ come from the prose following PPPB Eq. 19, not its magnetic-field Table 1.

FFM limits require matched microfields, damping, populations and Doppler settings. Motional narrowing also requires finite relevant moments; truncated and untruncated heavy-tailed fields cannot be interchanged. Binning is approximate: reduce bin tolerance relative to all relevant energies and compare against the unbinned result. Doppler is applied after the nonlinear FFM transformation here; ZEST's placement inside its construction of J is a different model and need not commute. No universal most-accurate G function, safe bin size or default core quadrature is established by the present tests.

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1 Introduction

StarkZee computes hydrogenic plasma line shapes when electric microfields and magnetic fields must be treated together rather than as independent corrections. The target problem is common in fusion spectroscopy, dense laboratory plasmas, and compact-object atmospheres: ionic microfields broaden and mix the emitter states through the Stark effect, while an imposed magnetic field splits and further mixes the same states through the Zeeman effect. When the two energy scales are comparable, a post-processed Stark profile with a Zeeman triplet added afterward is no longer sufficient.

The code therefore diagonalizes the Stark-Zeeman Hamiltonian directly at each sampled ionic microfield. It then averages the resulting polarized profiles over the microfield distribution and adds electron-impact broadening and optional ion dynamics through the Frequency Fluctuation Model (FFM). Thermal Doppler broadening can be included inside either solver; instrumental broadening is an explicit post-processing convolution.

This manual is organized from use to derivation. Section 2 gives the computational workflow and package map. Section 3 derives the line-shape equations and the atomic, plasma, and numerical ingredients. Later sections summarize validation features, built-in comparison models, and the physical limits of the approximations.

2 Calculation Workflow and Package Map

The public entry point is the `LineProfile` class. It converts wavelength, frequency, wavenumber, or energy grids to photon energy, delegates to the explicitly selected solver, stores the polarization profiles and corresponding spectral axes, and combines the components for the requested observation angle. `compute_profile` is the historical name of the quasi-static path and is an alias of `compute_static_profile`; `compute_ffm_profile` is the separate dynamic-ion path and requires `Ti_ev`. The code does not switch between these models automatically.

Use `compute_discrete(Fz,Fx)` for unbroadened transition sticks at one specified electric-field configuration. For a continuous plasma profile, use the static solver when the ion fluctuation energy is small compared with the characteristic Stark-dressed-transition spread, and FFM when ion motion changes the line core. If the separation is unclear, run both on the same grid with the same quadrature, microfield, electron, and Doppler settings. Density alone is not a universal boundary because it changes both the microfield distribution and the fluctuation rate.

Internally, the high-level calls delegate the work through a small number of focused modules:

1. `line_profile.py` manages spectral-grid conversion, result storage, detuning axes, and the Stokes observation-angle combination; it does not normalize profiles or apply the instrumental slit.
2. `static_profile.py` performs the quasi-static microfield average and can include Doppler broadening when `Ti_ev` is supplied.
3. `radiator.py` constructs the field-free plus magnetic radiator Hamiltonian H_A and the transition dipole matrices in the uncoupled $|n, l, m_l, s, m_s\rangle$ basis.

4. `solve_starkzee()` forms the complete Hamiltonian $H = H_A + V_E$ at each microfield point and diagonalizes it as one matrix. The Stark term is not added perturbatively.
5. `broadening.py` evaluates electron-impact widths, including the frequency-dependent Griem-Baranger-Kolb (GBK) model and the ZEST-inspired variant.
6. `ffm.py` collects the field-dressed transitions and replaces the frozen-field average with Markov mixing at the ion fluctuation rate; its Doppler convolution is enabled by default.
7. `microfield.py` supplies Hooper, Holtsmark, Potekhin, and custom tabulated microfield distributions.
8. `convolutions.py` provides standalone wavelength-space Doppler and instrumental broadening helpers; instrumental broadening is always explicit post-processing.
9. `models/` contains comparison models: `voigt`, `lomanowski`, `stehle_param`, `stehle`, and `rosato`.

The numerical calculation has one central loop: for each sampled microfield magnitude F and orientation $\mu = \cos \theta$, StarkZee builds $V_E(F, \mu)$, diagonalizes the upper and lower manifolds, and rotates the transition dipoles into the dressed-state bases. The resulting transition energy and strength define one Stark-Dressed Transition (SDT). The dressed states are therefore calculated inside the microfield loop, before broadening or ion-dynamic mixing. The static solver broadens and accumulates each SDT immediately; FFM first collects the SDTs from all field configurations and then mixes them through the fluctuation process.

3 Physics and Numerical Formulation

3.1 Liouville Space Representation

The spectral line-shape intensity $I(\omega)$ of emitted radiation is related to the Fourier transform of the dipole autocorrelation function $C(t)$ by [2]:

$$I(\omega) = \frac{1}{\pi} \text{Re} \int_0^\infty C(t) e^{i\omega t} dt \quad (1)$$

In Liouville space notation, the autocorrelation function is a trace over the quantum emitter states:

$$C(t) = \langle \langle \vec{d}^* | U(t) | \vec{d} \rho_0 \rangle \rangle \quad (2)$$

where $\vec{d}$ is the electric transition dipole operator, ρ_0 is the equilibrium density matrix of the initial manifold, and $U(t)$ is the time-evolution propagator.

In the optical-coherence subspace of operators $|i_u\rangle\langle j_l|$, H_u acts on the ket/upper index and H_l on the bra/lower index. The dual operator is the transpose for this vectorization convention, not an additional independent conjugation. The no-quenching Liouvillian is:

$$L = \frac{1}{\hbar} (H_u \otimes \hat{I}^d - \hat{I} \otimes H_l^d) \quad (3)$$

In a product of manifold eigenbases, L is diagonal with eigenvalues $(E_i^u - E_j^l)/\hbar$. In the uncoupled basis it also contains off-diagonal Hamiltonian couplings.

From propagator to resolvent. The time-evolution propagator is generated by L :

$$U(t) = e^{-iLt} \quad (4)$$

Substituting into Eq. (2) gives the explicit correlation function

$$C(t) = \langle \langle \vec{d}^* | e^{-iLt} | \vec{d} \rho_0 \rangle \rangle \quad (5)$$

Inserting this into Eq. (1) and evaluating the one-sided Fourier transform converts the time-domain exponential into a frequency-domain resolvent:

$$I(\omega) = \frac{1}{\pi} \text{Re } i \langle \langle \vec{d}^* | (\omega \hat{I} - L + i0^+)^{-1} | \vec{d} \rho_0 \rangle \rangle \quad (6)$$

Fast electron collisions are encoded as an additional imaginary damping operator $\Phi(\omega)$ in the same Liouville space (Section 3.8), shifting $L \rightarrow L - i\Phi(\omega)$ ($\Phi(\omega) > 0$ giving decay, consistent with the $+i\Phi(\omega)$ appearing inside the resolvent below):

$$I(\omega) = \frac{1}{\pi} \text{Re } i \langle \langle \vec{d}^* | (\omega \hat{I} - L + i\Phi(\omega))^{-1} | \vec{d} \rho_0 \rangle \rangle \quad (7)$$

Field-dependent Liouvillian and static profile. Under the quasi-static ion approximation the radiator is exposed to a static ionic microfield $\vec{F}$ at angle θ to $\vec{B}$. Both manifold Hamiltonians H_u and H_l then depend on the field magnitude F and orientation $\mu = \cos \theta$ through the Stark perturbation V_E (Section 3.4), making the Liouvillian parametrically field-dependent:

$$L(F, \mu) = \frac{1}{\hbar} [H_u(F, \mu) \otimes \hat{I}^d - \hat{I} \otimes H_l^d(F, \mu)], \quad H(F, \mu) = H_A + V_E(F, \mu) \quad (8)$$

where H_A is the field-free magnetic Hamiltonian (Section 3.2) and $V_E(F, \mu)$ is the linear Stark interaction (Section 3.4). Equations (7) and (8) together give the field-dependent q -polarized profile at a given configuration (F, μ) :

$$I_q(\omega, F, \mu) = \frac{1}{\pi} \text{Re } i \langle \langle d_q^* | (\omega \hat{I} - L(F, \mu) + i\Phi(\omega))^{-1} | d_q \rho_0 \rangle \rangle \quad (9)$$

The total q -polarized profile is then the average over the ionic microfield distribution $W(F)$ and orientation:

$$I_q(\omega) = \frac{1}{2} \int_0^\infty \int_{-1}^1 W(F) I_q(\omega, F, \mu) d\mu dF \quad (10)$$

Because reflection in the plane perpendicular to the axial magnetic field maps the two signs of μ while preserving squared dipoles and the assumed populations/damping, $I_q(\omega, F, \mu) = I_q(\omega, F, -\mu)$. The matrix itself is not even in μ ; consequently the integral over $[-1, 1]$ reduces to twice the $[0, 1]$ range, discretized by Gauss–Legendre quadrature. The observed intensity at angle α between the line of sight and $\vec{B}$ is:

$$I(\omega, \alpha) = I_\pi(\omega) \sin^2 \alpha + \frac{1}{2} (I_{\sigma^+}(\omega) + I_{\sigma^-}(\omega)) (1 + \cos^2 \alpha) \quad (11)$$

Sections 3.2–3.8 detail each ingredient of Eq. (9); Section 3.10 extends this framework to include ion dynamics via the FFM.

3.2 Radiator Hamiltonian in Magnetic Fields

For each principal quantum number shell n , the atomic Hamiltonian H_A is constructed in the canonical uncoupled hydrogenic basis $|n, l, m_l, s=\frac{1}{2}, m_s\rangle$ of dimension $2n^2$. Basis states are ordered by an outer loop over $l \in [0, n-1]$, an intermediate loop over $m_l \in [-l, l]$, and an inner loop over $m_s \in \{+\frac{1}{2}, -\frac{1}{2}\}$. Five physical contributions are assembled into a single matrix [3]:

$$H_A = H_0 + H_{\text{SO}} + H_{\text{FS}} + H_Z^{(1)} + H_Z^{(2)} \quad (12)$$

3.2.1 Unperturbed Hydrogenic Energy (H_0)

The diagonal unperturbed energy is corrected for the finite nuclear mass:

$$\langle n, l, m_l, m_s | H_0 | n, l, m_l, m_s \rangle = -\frac{Z^2 \text{Ry}_{\text{red}}}{n^2}, \quad \text{Ry}_{\text{red}} = \text{Ry}_{\infty} \frac{M_{\text{nuc}}}{m_e + M_{\text{nuc}}} \quad (13)$$

where M_{nuc} is the nuclear mass of the emitting isotope (fixed by its atomic mass number) and Ry_{red} the corresponding reduced-mass Rydberg constant. When `use_empirical_data=True` this analytic diagonal is replaced by NIST tabulated energies (Section 3.3).

3.2.2 Spin-Orbit Coupling (H_{SO})

The spin-orbit interaction $H_{\text{SO}} = \zeta_{nl} \vec{L} \cdot \vec{S}$ couples orbital and spin angular momenta. The coupling constant for $l > 0$ is:

$$\zeta_{nl} = \frac{Z^4 \alpha^2 \text{Ry}_{\infty}}{n^3 l (l + \frac{1}{2}) (l + 1)} \quad (14)$$

Using the ladder decomposition $\vec{L} \cdot \vec{S} = L_z S_z + \frac{1}{2}(L_+ S_- + L_- S_+)$, the matrix elements within a given l -subshell are:

- **Diagonal** ($\Delta m_l = 0, \Delta m_s = 0$):

$$\langle l, m_l, m_s | H_{\text{SO}} | l, m_l, m_s \rangle = \zeta_{nl} m_l m_s \quad (15)$$

- **Off-diagonal** ($\Delta m_l = \pm 1, \Delta m_s = \mp 1$): for $s = \frac{1}{2}$ the spin ladder factor equals 1 for all allowed transitions:

$$\langle l, m_l \pm 1, m_s \mp 1 | H_{\text{SO}} | l, m_l, m_s \rangle = \frac{1}{2} \zeta_{nl} \sqrt{l(l+1) - m_l(m_l \pm 1)} \quad (16)$$

3.2.3 Fine-Structure Relativistic Corrections (H_{FS})

To restore the Dirac degeneracy (e.g. $2s_{1/2}$ and $2p_{1/2}$), the mass-velocity and Darwin terms are added [3]. These are diagonal in $|n, l, m_l, m_s\rangle$ and independent of m_l and m_s :

$$\langle l | H_{\text{FS}} | l \rangle = \begin{cases} -A_{\text{FS}}(n - \frac{3}{4}) & l = 0 \\ -A_{\text{FS}}\left(\frac{n}{l + \frac{1}{2}} - \frac{3}{4}\right) & l > 0 \end{cases} \quad (17)$$

where $A_{\text{FS}} = Z^4 \alpha^2 \text{Ry}_{\infty} / n^4$. The algebraic identity $H_{\text{SO}} + H_{\text{FS}}$ reproduces the Dirac fine-structure eigenvalues through the retained order $O((Z\alpha)^4)$ for both $j = l \pm \frac{1}{2}$.

3.2.4 Linear Zeeman Effect ($H_Z^{(1)}$)

$$\langle m_l, m_s | H_Z^{(1)} | m_l, m_s \rangle = \mu_B B (m_l + g_s m_s) \quad (18)$$

where $\mu_B \approx 5.788 \times 10^{-5} \text{ eV/T}$ and $g_s \approx 2.0023192$.

3.2.5 Quadratic (Diamagnetic) Zeeman Effect ($H_Z^{(2)}$)

$$H_Z^{(2)} = \frac{e^2 B^2}{8m_e} r^2 \sin^2 \theta \quad (19)$$

This operator couples states with $\Delta l = 0$ and $\Delta l = \pm 2$. The matrix elements factor into radial and angular parts:

$$\langle l_1, m_l | H_Z^{(2)} | l_2, m_l \rangle = \left(\frac{e^2 B^2 a_0^2}{8m_e} \right) \langle n, l_1 | r^2 | n, l_2 \rangle \langle l_1, m_l | \sin^2 \theta | l_2, m_l \rangle \quad (20)$$

1. Radial integrals:

- Diagonal ($l_1 = l_2 = l$): $\langle n, l | r^2 | n, l \rangle = \frac{n^2}{2Z^2} [5n^2 + 1 - 3l(l+1)] [a_0^2]$
- Off-diagonal ($l_2 = l_1 \pm 2$): evaluated by a closed form by default, with optional numerical validation of $\int_0^\infty R_{nl_1} r^4 R_{nl_2} dr$, cached with `lru_cache`.

2. Angular integrals:

- Diagonal: for $l = 0$, $\langle \sin^2 \theta \rangle = 2/3$; for $l > 0$,

$$\langle l, m_l | \sin^2 \theta | l, m_l \rangle = 1 - \frac{2l^2 + 2l - 1 - 2m_l^2}{(2l-1)(2l+3)} \quad (21)$$

- Off-diagonal ($l_2 = l_1 + 2$, $l_{\text{low}} = \min(l_1, l_2)$): the constant term in $1 - \cos^2 \theta$ vanishes for orthogonal harmonics, leaving:

$$\langle l_1, m_l | \sin^2 \theta | l_2, m_l \rangle = -\sqrt{\frac{[(l_{\text{low}} + 1)^2 - m_l^2] [(l_{\text{low}} + 2)^2 - m_l^2]}{(2l_{\text{low}} + 1)(2l_{\text{low}} + 3)^2(2l_{\text{low}} + 5)}} \quad (22)$$

3.3 Empirical Field-Free Energies

Static profiles, FFM profiles, and discrete transitions default to `use_empirical_data=True`. High-level calls select the emitting isotope; direct solver calls must supply the matching atom. Use `use_empirical_data=False` for analytical energies, shells outside the bundled tables, or hydrogen-like ions with $Z > 1$. Low-level Hamiltonian functions retain their analytical default and eV units. Absent shell-average rows are derived from complete fine-structure tables using weights $2j + 1$, including tritium and the deuterium ground state. Incomplete shells raise an error. The analytic diagonal $H_0 = -Z^2 \text{Ry}_{\text{red}}/n^2$, together with the spin-orbit and fine-structure terms, reproduces the field-free level positions only to the accuracy of the hydrogenic Dirac formula. For quantitative line centers, StarkZee can instead inject *measured* field-free energies through the `use_empirical_data=True` flag (with the element symbol atom). The values are read from a tabulated database (`atomic_data.load_levels`) holding NIST level energies [16] in wavenumbers [cm^{-1}], resolved by (n, l, j) . The database is stored in `starkzee/data/atomic_levels.json`; it currently holds hydrogen ("H", $n \leq 8$), deuterium ("D", $n \leq 6$), and tritium ("T", $n \leq 3$). Each top-level key is an atom symbol; the value is an object with two sections: "fine_structure_true" (entries with fields `n`, `l`, `j`, `energy`) and "fine_structure_false" (entries with fields `n`, `energy`). All energies are in cm^{-1} above the ground state (NIST vacuum values). Additional atoms can be supported by adding the corresponding entry to this file. To refresh the bundled data from the NIST ASD levels query, run `python scripts/update_nist_levels.py H -spectrum "H I"`. The updater is a development

tool: it writes the local JSON database and checks that each requested fine-structure shell contains all hydrogenic (l, j) states before replacing the data; it also keeps NIST rows marked uncertain (a trailing ? on the energy value) rather than silently dropping them. The empirical Hamiltonian is kept in cm^{-1} throughout this code path so that callers can work entirely in wavenumber units; all Zeeman contributions are also converted from eV to cm^{-1} before being added, keeping every term on the same scale.

Beyond the bare Hamiltonian. Both solvers forward empirical data and isotope selection. Stark matrices are converted to inverse centimetres before addition to empirical Hamiltonians. The field-dependent centering shift is used only to reconstruct eigenvalues; physical broadening uses the fixed zero-field reference. High-level calls default to the emitting isotope; low-level callers must set atom explicitly.

Because the field-free Hamiltonian is degenerate (the Dirac formula makes $2s_{1/2}$ and $2p_{1/2}$ coincide), the empirical energies cannot simply be written on the diagonal of the uncoupled $|n, l, m_l, m_s\rangle$ basis: `numpy.linalg.eigh` is free to mix the degenerate l states arbitrarily. StarkZee therefore:

1. diagonalizes a degeneracy-broken field-free Hamiltonian (unperturbed energy + spin-orbit only, *omitting* the mass-velocity/Darwin term so that l remains a good label), yielding eigenvectors V ;
2. labels each coupled eigenstate k by its dominant orbital component l and its total angular momentum: for $l = 0, j = 1/2$; for $l > 0$, the branch $j = l \pm 1/2$ is set by the sign of the spin-orbit expectation value. The coefficient ξ_{nl} is defined only for $l > 0$;
3. assigns the tabulated energy $D_k = E^{\text{emp}}(l, j)$ [cm^{-1}] to each eigenstate and reconstructs the field-free Hamiltonian as

$$H_0^{\text{emp}} = V \text{diag}(D) V^\dagger \quad [\text{cm}^{-1}]. \quad (23)$$

The Zeeman terms ($\mu_B B(m_l + g_s m_s)$ and the diamagnetic correction) are converted from eV to cm^{-1} and added on top of H_0^{emp} . Since the tabulated values are absolute level energies (ground state at 0), transition wavenumbers follow directly as differences $E_u - E_l$ and reproduce the selected tabulated level differences, not necessarily observed intensity centroids.

Result references. The analytic E0 remains available for comparison. Result detunings and Doppler/electron broadening use a fixed zero-field shell-trace reference in empirical mode, distinct from the magnetic-field-dependent shift used only to condition and reconstruct eigenvalues. Empirical metadata are cleared on analytic recomputation. A shell reference is not an observed intensity centroid.

3.4 Full Electron-Radiator Hamiltonian

Under the quasi-static ion approximation, the radiator is subjected to a constant electric microfield $\vec{F}$ at an angle θ relative to the magnetic field $\vec{B}$. The microfield is decomposed into longitudinal ($F_z = F \cos \theta$) and transverse ($F_x = F \sin \theta$) components. The Stark interaction is

$$V_E = -e \vec{r} \cdot \vec{F} = -e(zF_z + xF_x) = -\hat{d} \cdot \vec{F} \quad (24)$$

where $\hat{\mathbf{d}} = e\vec{r}$ is the electric dipole operator. The full electron-radiator Hamiltonian is

$$H = H_A + V_E = H_0 + H_{\text{SO}} + H_{\text{FS}} + H_Z^{(1)} + H_Z^{(2)} + V_E \quad (25)$$

H_A and V_E are assembled into a single matrix and diagonalized *as a whole* by `numpy.linalg.eigh` at every microfield quadrature point (`solve_starkzee`). The Stark interaction is **not** treated perturbatively: there is no expansion in powers of V_E . This exact treatment is essential when the Stark energy $eF\langle r \rangle_n$ is comparable to the fine-structure or Zeeman splittings, yielding the Stark-dressed eigenstates $|k\rangle$ and transition frequencies ω_k .

3.4.1 Stark Perturbation Matrix Elements

The restriction $\Delta n = 0$ is a basis truncation, not an electric-dipole selection rule. Within that truncated basis the dipole rules are $|l_i - l_j| = 1$ and $\Delta m_s = 0$. In eV:

$$\langle n, l_i, m_{l,i} | V_E | n, l_j, m_{l,j} \rangle = -\delta_{m_{s,i}, m_{s,j}} \langle n, l_i | r | n, l_j \rangle \langle l_i, m_{l,i} | (zF_z + xF_x) / r | l_j, m_{l,j} \rangle \quad (26)$$

Radial matrix element. For $l = \max(l_i, l_j)$, Gordon's formula gives the within-shell radial dipole element exactly:

$$\langle n, l | r | n, l-1 \rangle = \frac{3n}{2Z} \sqrt{n^2 - l^2} \quad [a_0] \quad (27)$$

Angular matrix elements. The angular factor is the spherical dipole tensor, written $z/r = T_0^{(1)}$ and $x/r = (T_{-1}^{(1)} + T_{+1}^{(1)}) / \sqrt{2}$. Its non-zero matrix elements (with $l = \max(l_1, l_2)$) are:

$$\langle l-1, m_l | T_0^{(1)} | l, m_l \rangle = \sqrt{\frac{l^2 - m_l^2}{(2l-1)(2l+1)}} \quad (q=0, \Delta m_l=0) \quad (28)$$

$$\langle l_1, m_{l,1} | T_{+1}^{(1)} | l_2, m_{l,2} \rangle = \begin{cases} +\sqrt{\frac{(l-m_{l,2})(l-m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ -\sqrt{\frac{(l+m_{l,2})(l+m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \quad (q=+1, \Delta m_l=-1) \quad (29)$$

$$\langle l_1, m_{l,1} | T_{-1}^{(1)} | l_2, m_{l,2} \rangle = \begin{cases} -\sqrt{\frac{(l+m_{l,2})(l+m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ +\sqrt{\frac{(l-m_{l,2})(l-m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \quad (q=-1, \Delta m_l=+1) \quad (30)$$

These same three matrix elements reappear unchanged in the transition dipole operator (Section 3.5), since both the Stark coupling and the radiative dipole are built from the rank-1 tensor $T_q^{(1)}$. The Stark matrices M_z and M_x [eV/(V/m)] are precomputed once per (n, Z) and cached (`_stark_templates`), so that $V_E = F_z M_z + F_x M_x$ is formed at each quadrature point by a simple scalar multiplication.

3.5 Transition Dipole Operator

The transition dipole matrices D_q ($q \in \{0, \pm 1\}$) are built in the uncoupled basis $|n, l, m_l, s, m_s\rangle$. Selection rules require $\Delta l = \pm 1$, $\Delta m_l = q$, and $\Delta m_s = 0$:

$$\langle n_l, l_l, m_{l,l}, m_{s,l} | D_q | n_u, l_u, m_{l,u}, m_{s,u} \rangle = \delta_{m_{s,l}, m_{s,u}} \langle n_l, l_l | r | n_u, l_u \rangle \langle l_l, m_{l,l} | T_q^{(1)} | l_u, m_{l,u} \rangle \quad (31)$$

3.5.1 Angular Dipole Matrix Elements

The angular factors $\langle l_l, m_{l,l} | T_q^{(1)} | l_u, m_{l,u} \rangle$ follow from the Wigner-Eckart theorem [5]. They are exactly the rank-1 spherical-tensor matrix elements already written for the Stark coupling in Eqs. (28)–(30), now evaluated between the lower and upper manifolds: $q = 0$ gives the π component ($\Delta m_l = 0$), $q = +1$ the σ^- component ($\Delta m_l = -1$), and $q = -1$ the σ^+ component ($\Delta m_l = +1$).

3.5.2 Hydrogenic Radial Dipole Integrals (Gordon's Formula)

For $n_u \neq n_l$, the inter-shell radial elements $\langle n_l, l_l | r | n_u, l_u \rangle$ are evaluated exactly using Gordon's (1929) analytical formula [4]:

Here (n_b, L) is the state with larger orbital angular momentum and $(n_s, L - 1)$ the other state; these labels are not energy ordering. In particular n_b may be smaller than n_s (e.g. 3s–2p).

$$\begin{aligned} \langle n_s, L - 1 | r | n_b, L \rangle = & \frac{1}{Z} \frac{(-1)^{n_s-L}}{4(2L-1)!} \sqrt{\frac{(n_b+L)!(n_s+L-1)!}{(n_b-L-1)!(n_s-L)!}} \\ & \times \frac{(4n_b n_s)^{L+1} (n_b - n_s)^{n_b+n_s-2L-2}}{(n_b + n_s)^{n_b+n_s}} \left[F_1 - \left(\frac{n_b - n_s}{n_b + n_s} \right)^2 F_2 \right] \end{aligned} \quad (32)$$

where $L = \max(l_u, l_l)$, and F_1, F_2 are terminating Gauss hypergeometric functions:

$$F_1 = {}_2F_1 \left(-n_b + L + 1, -n_s + L, 2L, -\frac{4n_b n_s}{(n_b - n_s)^2} \right) \quad (33)$$

$$F_2 = {}_2F_1 \left(-n_b + L - 1, -n_s + L, 2L, -\frac{4n_b n_s}{(n_b - n_s)^2} \right) \quad (34)$$

Because the first argument in both hypergeometric functions is a non-positive integer, each series terminates as a finite sum:

$${}_2F_1(-m, b, c, x) = \sum_{k=0}^m \frac{(-m)_k (b)_k}{(c)_k k!} x^k \quad (35)$$

where $(-m)_k$ is the Pochhammer symbol. This exact evaluation carries no integration cutoff error and is cached for reuse.

3.6 Diagonalization and Transition Dipole Basis Rotation

At each quadrature point (F, μ) the total Hamiltonians for both manifolds are:

$$H_u(F, \mu) = H_{A,u} + V_{E,u}(F_z, F_x), \quad H_l(F, \mu) = H_{A,l} + V_{E,l}(F_z, F_x) \quad (36)$$

Diagonalization via `numpy.linalg.eigh` gives eigenvectors V_u, V_l and eigenvalues E_u^p, E_l^k :

$$V_u^\dagger H_u V_u = \text{diag}(E_u^p), \quad V_l^\dagger H_l V_l = \text{diag}(E_l^k) \quad (37)$$

The transition energy for each dressed-state pair ($p \rightarrow k$) is $E_{pk} = E_u^p - E_l^k$. The uncoupled dipole matrices D_q are rotated into the mixed eigenstate bases:

$$D'_q = V_l^\dagger D_q V_u, \quad S_q(p, k) = |D'_q(k, p)|^2 \quad (38)$$

The weight $S_q(p, k)$ and energy E_{pk} together define one Stark-Dressed Transition (SDT). The static profile at *one fixed* field configuration (F, μ) — Eq. (9) — is then realized in practice as a sum over all SDTs at that configuration, in energy units [eV] to match the returned spectral axis:

$$I_q(E, F, \mu) = \sum_{p,k} S_q(p, k) \frac{\gamma_e / \pi}{(E - E_{pk})^2 + \gamma_e^2} \quad (39)$$

No quadrature weight appears here: W_{ij} only multiplies this conditional profile once, in the outer microfield average of Eq. (10)/Eq. (70) – including it here too would double-count the field probability. where γ_e is the electron-impact half-width of Section 3.8.

3.7 Plasma Microfield Distributions

The electric microfield is averaged over a probability distribution $W(F)$. The presence of $\vec{B}$ breaks spherical symmetry, so the integration must be performed over both the field magnitude $F = |\vec{F}|$ and the orientation $\mu = \cos \theta$ between $\vec{F}$ and $\vec{B}$.

3.7.1 The Hooper Screened Microfield Distribution

To account for Debye screening of the ion Coulomb fields by surrounding plasma electrons, StarkZee implements a Hooper-like screened microfield distribution [12]:

$$W(\beta, a) = \frac{2\beta}{\pi} \int_0^\infty y \sin(\beta y) T(y, a) dy \quad (40)$$

Provenance note. This is the analytic screened characteristic-function ansatz below — it reduces to Holtsmark at $a = 0$ and approximates the Debye suppression Hooper’s low-frequency-component theory predicts — not an interpolation of Hooper’s own tabulated numerical results.

Known validity issue. This ansatz’s raw (unclipped) inverse transform is negative over part of the beta range once a is large enough – confirmed numerically ($a = 1, \beta \approx 5$ gives roughly -0.025) and, for the unregularized formula only, by a characteristic-function argument: near $t = 0$ the screened radial characteristic function satisfies $(1 - \text{Re } T(t))/t^2 \rightarrow 0$, so by Fatou’s lemma the corresponding field component’s second moment would have to be exactly zero – i.e. this cannot be the characteristic function of any non-degenerate field distribution, for any $a > 0$. `hooper_distribution` clips negative values to zero; `microfield_quadrature` subsequently normalizes the finite grid, which is a numerically convenient patch rather than a validated correction, and now raises a `UserWarning` above $a \approx 0.3$ (where the effect becomes numerically visible) to make this explicit. For quantitative work at strong screening prefer a tabulated distribution via `custom_table_path`, or the Potekhin distribution below.

Neither of the two reference codes this project otherwise tracks (Section 3.8’s ZEST and PPPB conventions) actually uses this ansatz: ZEST (Gilleron & Pain 2018) uses the Potekhin, Chabrier & Gilles (2002) fit exclusively for its microfield, and PPPB (Ferri, Peyrusse & Calisti 2022) uses the APEX model [13, 14] for the general case and the *genuine* Hooper (1968) distribution – not this closed-form stand-in – specifically for neutral emitters. See TODO item 2 for the full comparison.

The all-positive-screening proof above is restricted to the unregularized expression. The implemented $y^2 + 10^{-8}$ denominator changes the small- y limit, so failure of that implementation is established numerically, not by that proof. **Selecting a distribution.** Screened mode now defaults to Potekhin, not the invalid Hooper-like ansatz. If ion

temperature is omitted in default mode, a warning announces the equal-temperature assumption. Explicit Potekhin selection requires ion temperature. Custom tables, explicit emitter charge, charged-point selection and background ion charge are forwarded by both solvers. For a hydrogen-like ion the emitter charge defaults to $Z - 1$; the current Potekhin fit uses only neutral versus charged status, not the charge magnitude. Invalid custom data raise rather than silently falling back.

FFM rate conventions. The default ZEST convention uses the most-probable perturber speed $\sqrt{2kT/m_p}$. The optional PPP-manual convention uses $\sqrt{kT/\mu}$ with the emitter-perturber reduced mass. The emitter and perturber masses are separate inputs; omitting the latter retains the historical same-species assumption. Both formulae coincide for equal masses. where:

- $\beta = F/F_0$ is the reduced electric field strength.
- F_0 is the **Holtmark normal field strength**, the Coulomb field of a single perturber at the mean inter-particle distance r_e :

$$F_0 = \frac{e}{4\pi\epsilon_0 r_e^2}, \quad r_e = \left(\frac{3}{4\pi N_e} \right)^{1/3} \quad (41)$$

- $a = r_e/\lambda_D$ is the Debye screening parameter, where λ_D is the multi-species Debye length:

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{N_e e^2 (1 + \sum_i X_i Z_i^2)}} \quad (42)$$

with $X_i = N_i/N_e$ the fractional ion concentration. This uses a single temperature T_e for every species' contribution to the screening sum – valid when $T_i \approx T_e$ (an electron-only response instead omits the ion sum); it is not a general multi-temperature ($T_i \neq T_e$) Debye length.

- $T(y, a) = \exp(-y^{3/2} S(y, a))$ is the screened characteristic function, with Debye screening factor:

$$S(y, a) = \left(1 + \frac{c a^2}{y^2} \right)^{-3/4}, \quad c = \begin{cases} 1.5 & \text{charged radiator} \\ 1.0 & \text{neutral radiator} \end{cases} \quad (43)$$

Setting $a = 0$ yields $S = 1$ and $T = \exp(-y^{3/2})$, recovering the unscreened **Holtmark distribution** $W_H(\beta)$ [11].

3.7.2 The Potekhin Microfield Distribution

The analytical fits of Potekhin et al. (2002) [15] describe the cumulative distribution $Q(\beta)$ and probability density $P(\beta)$ for both screened ($s > 0$) and unscreened ($s = 0$) potentials, for neutral and charged radiators, incorporating the ion-ion coupling parameter Γ :

$$Q(\beta) = \frac{q_0 \beta^3 - 1.33 \beta^{9/2} + \beta^6}{q_1 + q_2 \beta^2 + q_3 \beta^3 - \frac{1}{3} \beta^{9/2} + \beta^6} \quad (s = 0) \quad (44)$$

$$P(\beta) \approx \beta^2 S_N \left[A e^{-a\beta^\alpha} + B e^{-b\beta^\gamma} + \frac{e^{-\Gamma\beta^{1/2}}}{1 + c\beta^{9/2}} \right] \quad (s > 0) \quad (45)$$

where $q_n, A, a, \alpha, B, b, \gamma, c$ are tabulated functions of Γ and s , and S_N is the normalization constant.

3.7.3 Quadrature Discretization

The double integral $\iint W(F, \mu) dF d\mu$ (Eq. 10) is discretized as follows:

1. **Field magnitude F :** A uniform grid of N_F points (`num_f`) in reduced field space $\beta_i \in [0, \beta_{\max}]$ (`max_beta`, default $\beta_{\max} = 10$):

$$F_i = \beta_i F_0, \quad W_{F,i} = \frac{W(\beta_i, a) \Delta\beta}{\sum_j W(\beta_j, a) \Delta\beta}, \quad \sum_i W_{F,i} = 1 \quad (46)$$

The core of $W(\beta)$ sits below $\beta \approx 5$ –8, but the Holtsmark tail decays only as $\beta^{-5/2}$: about 3 % of the probability lies beyond $\beta = 10$, and truncating it (the weights above are renormalized to sum to 1) removes exactly the strong-field configurations that build the quasi-static far wings of the line. Increase `max_beta` (together with `num_f`, since the same point count must now cover a wider range) for far-wing studies; line-core adequacy also requires quadrature convergence checks.

2. **Field orientation μ :** N_μ -point **Gauss-Legendre quadrature** over $[0, 1]$. Legendre roots $x_j \in [-1, 1]$ and weights w_j are mapped as:

$$\mu_j = \frac{x_j + 1}{2}, \quad W_{\mu,j} = \frac{w_j}{2} \quad (47)$$

3. **Combined weight:** $W_{ij} = W_{F,i} W_{\mu,j}$. Points with $W_{ij} < 10^{-15}$ are skipped.

so that at each active point the field decomposes into the longitudinal and transverse components seen by the radiator,

$$F_{z,ij} = F_i \mu_j, \quad F_{x,ij} = F_i \sqrt{1 - \mu_j^2}, \quad (48)$$

which are passed directly to the Stark template (Section 3.4). The weight $W(\beta_i, a)$ is drawn from whichever distribution the user selects — Holtsmark, Hooper, or Potekhin (above) — while the orientation quadrature is identical in every case.

3.8 Electron Impact Broadening

Fast-moving electrons are treated in the *impact* (completed-collision) approximation: each collision is completed in a time short compared to both the inverse linewidth and the mean inter-collision time. In this limit, successive collisions are statistically independent and their net effect on the line shape enters through a damping operator $\Phi(\omega)$ in the Liouville-space resolvent (Eq. 7).

3.8.1 Liouville-space structure of $\Phi(\omega)$ and connection to $\langle r^2 \rangle$

In the no-quenching approximation (electron collisions do not mix the upper and lower radiating manifolds), StarkZee treats the operator Φ in Liouville space as a sum of *independent* contributions from each manifold. This is an additional approximation layered on top of no-quenching, not a consequence of it: absence of manifold-mixing transitions does not by itself rule out a *correlated* (common-mode) phase shift affecting both manifolds together, which would leave the transition frequency unchanged rather than broadening it — unlike independent dephasing of each level, which does broaden the line. The independent-dephasing and interference-neglect approximations should be understood as separate, additional assumptions:

$$\Phi(\omega) = \frac{1}{2} (\Phi_u(\omega) \otimes \hat{I}^d + \hat{I} \otimes \Phi_l^d(\omega)) \quad (49)$$

where Φ_u and Φ_l are self-energy operators acting within the upper (n_u) and lower (n_l) manifolds respectively.

The GBK semi-classical model [10, 8] evaluates Φ_n from the leading long-range term of the electron–radiator interaction, which is the *charge-dipole* coupling of the passing electron’s charge with the atomic dipole, $V_{ee} \propto \vec{r} \cdot \vec{r}_e / r_e^3$ (order r_{atom} / r_e^2) – not a dipole–dipole interaction. Squaring this interaction in the second-order broadening calculation is what produces the atomic-dipole-squared operator r_n^2 . Averaging over the Maxwell–Boltzmann distribution of electron velocities and over the transverse impact-parameter area element $2\pi\rho d\rho$, this yields a self-energy proportional to the operator r_n^2 within shell n :

$$\Phi_n(\omega) = W_0(\omega) r_n^2 [C_n + G(\omega)] \quad (50)$$

where r_n^2 is the mean-square-displacement operator within shell n , W_0 is the density-temperature prefactor (below), and $G(\omega)$ is the frequency-dependent GBK factor (below). The r^2 dependence is the direct quantum-mechanical trace of the dipole-coupling cross-section: a state with larger electronic extent couples more strongly to a passing electron.

Rotating the Hamiltonian does not diagonalize the collision operator. Independent SDT damping is an additional secular approximation, requiring off-diagonal collision couplings to be small relative to the relevant level gaps. The optional operator mode retains diagonal radius expectation values only; it does not implement the full collision superoperator. The illustrative upper contribution is:

$$\gamma_k^{(u)} = W_0 \langle k | r_u^2 | k \rangle [C_{n_u} + G(\Delta E_k)] \quad (51)$$

The resolvent $(\omega \hat{I} - L(F, \mu) + i\Phi(\omega))^{-1}$ then has eigenvalues $(\omega - \omega_k + i\gamma_k)^{-1}$ in the SDT basis, and $\text{Re}[i/(\omega - \omega_k + i\gamma_k)] = \gamma_k/[(\omega - \omega_k)^2 + \gamma_k^2]$, so Eq. (9) reduces directly to the positive, resonance-peaked Lorentzian sum of Eq. (70).

Full PPP impact-limit operator and interference. The optional `electron_interference=True` path follows Section 2.2 and Appendix B of Calisti, Ferri, Mossé and Talin, *The PPP Code – User Manual* (2024), HAL [hal-04501367v1](#). In the optical-coherence basis $|\alpha\beta\rangle\rangle$, its Eq. (B1) is

$$\begin{aligned} \Phi_{\alpha\alpha'\beta\beta'} &= \delta_{\beta\beta'} \sum_{\alpha''} \mathbf{d}_{\alpha\alpha''} \cdot \mathbf{d}_{\alpha''\alpha'} G + \delta_{\alpha\alpha'} \sum_{\beta''} \mathbf{d}_{\beta'\beta''} \cdot \mathbf{d}_{\beta''\beta} G \\ &\quad - \mathbf{d}_{\alpha\alpha'} \cdot \mathbf{d}_{\beta'\beta} (G + G). \end{aligned} \quad (52)$$

The first two terms are the upper- and lower-manifold self contributions; the third is the upper–lower interference term coupling distinct radiative coherences. StarkZee presently implements the manual’s impact-limit choice $G(\Delta\omega) = G(0)$: one upper-shell-calibrated coefficient multiplies all three terms, preserving the dissipative $\sum_c (R_{u,c} - R_{l,c})^2$ structure. The finite intermediate-state closure is the selected upper and lower principal shells.

For each ionic field configuration the code diagonalizes the general complex generator $K_f = L_f - i\Phi$ by a similarity rather than Hermitian eigendecomposition. Its left/right residues define $a_k + ic_k$, and the static component is

$$I_f(E) = \frac{1}{\pi} \sum_k \frac{a_k \gamma_k + c_k (E - \omega_k)}{(E - \omega_k)^2 + \gamma_k^2}. \quad (53)$$

Signed radiative dipoles are rephased into the same $(-1)^l$ convention used by the production Stark Hamiltonian before the complex residues are formed. `electron_interference=False`

remains the default. The full mode requires `frequency_dependent_width=False`; a point-wise $\Phi(E)$ would require a separate non-Hermitian solve at every observed energy. Its absolute accuracy and selected-shell closure still require a matched PPP/PPPB calculation.

The runnable `examples/model_comparison_non-hermit.py` script compares this full operator with the built-in models for D- α and D- γ . In the FFM, StarkZee's default experimental closure uses $p_k = |a_k| / \sum_j |a_j|$ as a nonnegative Markov distribution while retaining the signed $a_k + ic_k$ residue in the radiative numerator. This modulus rule is a documented StarkZee choice, not a rule recovered from the PPP manual. An opt-in grouping closure combines modes close in frequency and width, preserves their summed complex residue and first complex pole moment, and is accepted only if the grouped static profile meets a requested error bound, every grouped real residue is nonnegative, and every grouped width is positive.

Shell-average approximation. Computing the per-SDT matrix element $\langle k|r_u^2|k \rangle$ exactly requires rotating the r^2 operator to each dressed-state basis at every quadrature point (`electron_operator=True`). The default approximation replaces $\langle k|r_u^2|k \rangle$ by the shell average $\langle r^2 \rangle_{n_u}$, giving a single homogeneous half-width:

$$\gamma_e(\Delta E) = W_0 \langle r^2 \rangle_{n_u} [C_{n_u} + G(\Delta E)] \quad (54)$$

The lower-manifold contribution $\gamma_k^{(l)}$ is retained in principle through Eq. (49); in practice, for Balmer lines ($n_l = 2$, $\langle r^2 \rangle_2 = 33 a_0^2$ full shell average, or $13.5 a_0^2$ intra-shell) it is smaller than the $n_u = 3$ upper term ($\langle r^2 \rangle_3 = 153 a_0^2$ full, $81 a_0^2$ intra) and is dominated by the latter; the exact ratio also depends on the strong-collision constants C_n and the dynamical factor G , not on the radius alone.

3.8.2 Density-Temperature Prefactor

$$W_0 = \frac{4\pi}{3} N_e \sqrt{\frac{2m_e}{\pi k_B T_e}} \left(\frac{\hbar}{m_e} \right)^2 \frac{\hbar}{e} \quad [\text{eV } a_0^{-2}] \quad (55)$$

The $\hbar/e$ factor converts rad s^{-1} to eV.

3.8.3 Shell-Averaged Mean-Square Radius

$$\langle r^2 \rangle_n = \frac{1}{n^2} \sum_{l=0}^{n-1} (2l+1) \frac{n^2}{2Z^2} [5n^2 + 1 - 3l(l+1)] a_0^2 \quad (56)$$

This expression scales as n^4/Z^2 and is *not* replaced by an approximate constant.

Caution: which $\langle r^2 \rangle$ PPPB intends is ambiguous, and unresolved as of this writing. Ferri et al. (2022) Eq. (19) write the broadening operator as $\vec{R} \cdot \vec{R}$ with " $\vec{R}$ the (emitter) electron position operator operating in the subspace of PQN n ." Projecting $\vec{R}$ onto the shell before squaring gives the *intra-shell* sum $\langle r_{\text{intra}}^2 \rangle_n$ (below) — the same operator ZEST uses, for the same $\Delta n = 0$ reasons given in the ZEST-justification paragraph below (Section 3.8.6) — rather than the full closure sum above. StarkZee keeps the full form as the `electron_model='pppb'` default only because the package's Ferri-figure comparisons were produced with it; the projected reading is available as `electron_model='pppb-intra'` (or `r2_form='intra'` passed directly to `electron_impact_width`). The two differ by $\times 2.44$ at $n = 2$ and $\times 1.89$ at $n = 3$. To be settled against digitized Ferri width data.

3.8.4 Strong-Collision Constant

ZEST Section 2.2 [21] gives these values in prose and attributes the strong-collision addition to Griem, Blaha & Kepple (1979) [9]. The discussion surrounding that paper's Table I compares distorted-wave and semiclassical $2p/4p$ scattering cross sections and establishes the constants; the constants are not a table in ZEST itself. Ferri, Peyrusse & Calisti [1] repeat the same sequence in the prose immediately following Eq. (19); Ferri Table 1 instead concerns critical magnetic-field values. Here n is the radiative upper-state principal quantum number, so its domain begins at $n = 2$:

$$C_n = \begin{cases} 1.50 & n = 2 \\ 1.00 & n = 3 \\ 0.75 & n = 4 \\ 0.50 & n = 5 \\ 0.40 & n \geq 6 \end{cases} \quad (57)$$

3.8.5 Dynamical GBK Factor

$$G(\Delta E) = \frac{1}{2} E_1(y(\Delta E)), \quad y(\Delta E) = \left(\frac{n^2}{2Z} \right)^2 \frac{\Delta E^2 + E_c^2}{\text{Ry}_\infty T_e} \quad (58)$$

where ΔE [eV] is a generic detuning argument (see below for which reference energy is used where), $E_c = \hbar\omega_c$ [eV] is the cutoff energy, $\text{Ry}_\infty \approx 13.606$ eV is the Rydberg energy ($e^2/2a_0$, the ionization energy of hydrogen), and $E_1(x)$ is the first-order exponential integral function defined for $x > 0$ by:

$$E_1(x) = \int_x^\infty \frac{e^{-t}}{t} dt \quad (59)$$

Cutoff frequency. The collision integral diverges logarithmically at large impact parameters unless truncated at a maximum $\rho_{\max}$, or equivalently at a lower cutoff $\omega_c = v_{\text{th}}/\rho_{\max}$ [8]. StarkZee follows the formulation of Ferri, Peyrusse & Calisti [1]:

$$\omega_c = \max(\omega_p, \omega_e, \omega_L, \omega_{\alpha\alpha'}) \quad (60)$$

- $\omega_p = \sqrt{N_e e^2 / (\epsilon_0 m_e)}$ — **electron plasma frequency**: Debye screening suppresses interactions at $\rho > \lambda_D = v_{\text{th}}/\omega_p$.
- $\omega_e = v_{\text{th}}/r_e$ — **configuration-change frequency**: the impact approximation requires each collision to be completed before the surrounding electron configuration changes appreciably. At high density or low T_e this can be shorter than the Debye timescale (see Section 3.8.6).
- $\omega_L = eB/m_e$ — **electron Larmor frequency**: in $\vec{B}$, electrons follow helical trajectories, reducing the effective interaction duration.
- $\omega_{\alpha\alpha'}$ — **emitter level-splitting frequency**: zero for hydrogen (degenerate l -subshells).

The largest frequency dominates, imposing the tightest bound on $\rho_{\max}$.

Frequency-dependent vs. resonance-center width. Two different detunings appear in the profile and must not be conflated (both were previously written as a single ΔE): $\delta_{pk} = E - E_{pk}$ centers the Lorentzian denominator on each individual Stark-dressed transition (Eq. (70)), while $\delta_0 = E - E_0$ is the detuning from the *shared* gross-structure line center.

When `frequency_dependent_width=True` (default), γ_e is evaluated **pointwise at δ_0 , not δ_{pk}** – i.e. $\gamma_e(\delta_0)$ using the full $E_1(y(\delta_0))$ function, matching the PPPB convention $\Phi(\Delta\omega)$ of [1] Eq. (19), where $\Delta\omega$ is stated there as the detuning from line center. All Stark-dressed components share this single width function evaluated on the shared observation grid, so each component becomes non-Lorentzian: its far wings relax toward the strong-collision floor C_n as $G \rightarrow 0$, and a component sitting far from line center receives a reduced width evaluated at the shared detuning (not at its own δ_{pk} – the width and the Lorentzian centering use different detunings by design, matching the cited PPPB convention). Setting this flag to `False` fixes the width at the line-center value $\gamma_e(0)$, which reduces computation time at the cost of accuracy in the far wings.

The frequency-dependent width request is always honored. The legacy force flag is a compatibility no-op. FFM retains its separately documented constant-width model.

This single half-width $\gamma_e(\delta_0)$ is the only quantity the electron model hands to the profile builder; how it is applied to each Stark-dressed transition is described in Section 3.9.

3.8.6 ZEST electron broadening variant (`electron_model='zest'`)

Activating `electron_model='zest'` in `LineProfile.compute_profile` switches to the electron broadening model of the ZEST code [21]. The formula has the same structure as the default GBK expression but replaces the full shell-averaged $\langle r^2 \rangle_n$ with the *intra-shell* mean-square radius:

$$\gamma_e^{\text{ZEST}}(\Delta E) = W_{\text{pref}} \langle r_{\text{intra}}^2 \rangle_n [G_n + G_{\text{ZEST}}(\Delta E)] \quad (61)$$

Physical justification — $\Delta n = 0$ interaction channels. The ZEST paper (Section 2.1 of [21]) adopts two approximations that together restrict the electron broadening operator to within-shell matrix elements:

1. **No-quenching approximation:** perturbers may not induce transitions between the lower and upper manifolds participating in the radiator emission process.
2. **$\Delta n = 0$ interaction channels:** “only states belonging to the same Layzer complex [same principal quantum number n] may be mixed by Stark and/or Zeeman effects.”

A Layzer complex is the set of all n^2 states sharing principal quantum number n . These two restrictions reduce the formal sum over all intermediate states in the broadening operator to within-shell dipole matrix elements only, directly yielding $\langle r_{\text{intra}}^2 \rangle_n$ in place of the full $\langle r^2 \rangle_n$. The restriction is additionally *supported* (not made exact) by the GBK dynamical factor: an inter-shell detuning (e.g. the $n = 3 \rightarrow n = 2$ gap of ≈ 1.89 eV at the conditions of Fig. 2) strongly suppresses $G(\Delta\omega_{\text{inter}})$ as the detuning grows, so inter-shell contributions are small in that regime – but G does not vanish identically at any finite detuning, and the suppression weakens at higher temperature or smaller shell gaps (e.g. for high- n Rydberg states). Treat this as a model approximation to be checked quantitatively for the parameters at hand, not a proven exact zero.

Intra-shell squared radius. Starting from the intra-shell radial element (with the larger orbital angular momentum, $\max(l, l')$, always under the square root – writing it as a blanket ± 1 on the lower index alone is wrong for the downward branch) $\langle n, l | r | n, l + 1 \rangle = \frac{3n}{2Z} \sqrt{n^2 - (l + 1)^2}$, $\langle n, l | r | n, l - 1 \rangle = \frac{3n}{2Z} \sqrt{n^2 - l^2}$, and angular weight factors $C(l, l + 1) =$

$(l+1)/(2l+1)$, $C(l, l-1) = l/(2l+1)$, the per- l intra-shell sum is

$$r_{\text{intra},l}^2 = \frac{9n^2}{4Z^2} (n^2 - l(l+1) - 1) \quad [a_0^2] \quad (62)$$

This formula holds for all $l \in [0, n-1]$, including the boundary cases $l = 0$ (no $l-1$ neighbor) and $l = n-1$ (no $l+1$ neighbor). Averaging over the n^2 spatial states with $(2l+1)$ weight yields the exact closed form:

$$\langle r_{\text{intra}}^2 \rangle_n = \frac{9n^2(n^2-1)}{8Z^2} \quad [a_0^2] \quad (63)$$

Numerical values for H ($Z = 1$): $n = 2$: $13.5 a_0^2$; $n = 3$: $81 a_0^2$; $n = 4$: $270 a_0^2$. These are smaller than the full shell averages (33, 153, 468 a_0^2) by factors of 0.41, 0.53, and 0.58, respectively.

Maximum wave-number cutoff κ_m . The PPPB model fixes the minimum impact parameter as $\rho_{\min} = n^2 a_0 / (2Z)$ (Bohr orbit) and uses a combined cutoff $\omega_c = \max(\omega_p, \omega_e, \omega_L)$. The ZEST convention instead defines a temperature-dependent wave-number upper limit

$$\kappa_m = \min \left(\frac{Z}{n^2 a_0}, \frac{Z \sqrt{2m_e k_B T_e}}{\hbar n^2} \right) \quad (64)$$

The second branch is the inverse **Weisskopf radius** $\rho_W = n^2 \hbar / (Z m_e v_{\text{th}})$, $v_{\text{th}} = \sqrt{2k_B T_e / m_e}$ — the hydrogen strong-collision radius, *not* the thermal de Broglie length $\hbar / (m_e v)$. The geometric branch $Z / (n^2 a_0)$ dominates above $T_e \approx \text{Ry}_\infty \approx 13.6 \text{ eV}$; below that the Weisskopf branch takes over. The only cutoff frequency retained in all ZEST G-functions is the electron plasma frequency ω_p (no ω_e or ω_L terms), so B does not enter the ZEST broadening directly.

G-function choices (electron_model selector). Three alternatives are available for the dynamical factor $G(\Delta\omega)$ in Eq. (51), all sharing $\langle r_{\text{intra}}^2 \rangle_n$, the same $G_n = C_n$ strong-collision constants, and κ_m as defined above.

zest / zest-gbk (default ZEST). Closed-form exponential integral with a κ_m -derived argument:

$$G_{\text{GBK}}(\Delta\omega) = \frac{1}{2} E_1 \left(\frac{\Delta\omega^2 + \omega_p^2}{2x^2 \omega_p^2} \right), \quad x = \kappa_m \lambda_D \quad (65)$$

This has the same E_1 structure as the PPPB formula but with $x = \kappa_m \lambda_D = \lambda_D / \rho_{\min}$ (since $\kappa_m \approx 1/\rho_{\min}$) playing the role of $\lambda_D / \rho_{\min}$ — the reciprocal of $\rho_{\min} / \lambda_D$ — and only ω_p in the additive term.

zest-lee. Analytical *min*-blend of the impact ($\Delta\omega \rightarrow 0$) and wing ($\Delta\omega \rightarrow \infty$) limits:

$$\begin{aligned} G_0 &= \frac{1}{2} \left[\ln(1+x^2) - \frac{x^2}{1+x^2} \right], \\ G_\infty(\Delta\omega) &= \frac{1}{2} E_1 \left(\frac{\Delta\omega^2}{2x^2 \omega_p^2} \right), \\ G_{\text{Lee}}(\Delta\omega) &= \min(G_0, G_\infty(\Delta\omega)) \end{aligned} \quad (66)$$

G_0 (the line-center plateau) provides the plasma-cutoff ceiling (a min caps the result at G_0 , not a lower bound); the E_1 wing term contains only $\Delta\omega^2$ (no $+\omega_p^2$) because the regularization is already captured by G_0 . At large detuning the two forms agree.

zest-dufty. Full random-phase-approximation (RPA) numerical integration accounting for Landau damping:

$$G_{\text{Dufty}}(\Delta\omega) = \int_{\kappa_{\min}}^{\kappa_m} \frac{e^{-x^2}}{\kappa |\varepsilon(\kappa, \Delta\omega)|^2} d\kappa, \quad x = \frac{|\Delta\omega|}{\kappa v_{\text{th}}}, \quad v_{\text{th}} = \sqrt{\frac{2k_B T_e}{m_e}} \quad (67)$$

where the longitudinal RPA dielectric function is

$$\varepsilon(\kappa, \Delta\omega) = 1 + \frac{1}{(\kappa \lambda_D)^2} \left[1 - 2x \mathcal{D}(x) + i\sqrt{\pi} x e^{-x^2} \right] \quad (68)$$

and $\mathcal{D}(x) = e^{-x^2} \int_0^x e^{t^2} dt$ is the Dawson function. At zero detuning the dimensionless integrand reduces (with $u = \kappa \lambda_D$) to $u^3 / (1 + u^2)^2$, which vanishes as u^3 at $u \rightarrow 0$ – the $\kappa \rightarrow 0$ limit of this expression is finite, not divergent. The lower limit $\kappa_{\min} = |\Delta\omega| / (10 v_{\text{th}})$ is a practical numerical regularization of the quadrature (and itself vanishes at zero detuning), not a fix for a genuine infrared divergence. No universal accuracy ranking is established among the three models; this option evaluates one numerical quadrature per frequency point.

Agreement of the three G-functions must be tested for the chosen parameters; differences can appear in the far wings where Landau damping becomes significant.

3.9 Profile Accumulation

Each Stark-dressed transition uses the following phenomenological variable-width kernel. It is a Lorentzian only for constant width; neither unit integrated area nor the measured half-width follows when the width depends on observation energy:

$$L_{pk}(\Delta E) = \frac{1}{\pi} \frac{\gamma_e(\Delta E)}{\Delta E^2 + \gamma_e(\Delta E)^2} \quad (69)$$

The three polarization profiles are accumulated by vectorized summation over all microfield quadrature points and all SDTs:

$$P_q(E) = \sum_i W_{F,i} \sum_j W_{\mu,j} \sum_{p,k} S_q(p, k) L_{pk}(E - E_{pk}) \quad (70)$$

where $q=0 \rightarrow P_\pi$, $q=-1 \rightarrow P_{\sigma^+}$ (blue), $q=+1 \rightarrow P_{\sigma^-}$ (red). This direct summation realizes Eq. (10) in practice. The Lorentzian denominator is centered on each transition through $E - E_{pk}$, while the default frequency-dependent half-width of Section 3.8 is evaluated on the shared observation-detuning grid $\gamma_e(E - E_0)$. The three accumulated channels are the polarization profiles $I_\pi, I_{\sigma^+}, I_{\sigma^-}$ entering the observed intensity of Eq. (11); for a line of sight at angle α to $\vec{B}$ they combine as $P_\pi \sin^2 \alpha + \frac{1}{2}(P_{\sigma^+} + P_{\sigma^-})(1 + \cos^2 \alpha)$.

3.10 Frequency Fluctuation Model (FFM)

The profile assembled in Section 3.9 treats the ionic microfield as frozen during emission. When the ions move appreciably on the emission timescale, that quasi-static average must be replaced by a dynamic one. The FFM retains the same instantaneous atomic calculation but assembles its dressed transitions dynamically instead. Dynamic ion motion causes the static microfield to fluctuate over time; in the FFM [6] this is modeled as a stationary Markovian process that mixes the Stark-dressed transition components at rate ν_i .

The Stark-Zeeman Hamiltonian $H = H_A + V_E$ is still diagonalized at each field configuration (F, μ) , producing dressed-state frequencies ω_k and dipole weights $|d_k|^2$ — the Stark-Dressed Transitions (SDTs). This step is purely static: ω_k and $|d_k|^2$ depend only on the instantaneous ion field. Fast electron collisions add a homogeneous Lorentzian half-width γ_k to each SDT (the GBK width from Section 3.8), acting on a timescale short enough that the ion configuration does not change during a single collision. Ion dynamics enter exclusively through v_i , estimated as the inverse time for an ion to cross the mean ion spacing at its thermal speed:

$$v_i = \frac{v_{th}}{r_i}, \quad v_{th}^{ZEST} = \sqrt{\frac{2k_B T_i}{M_{pert}}}, \quad r_i = \left(\frac{3}{4\pi N_i} \right)^{1/3} \quad (71)$$

This is the default ZEST convention. The optional PPP-manual convention uses

$$v_{th}^{PPP} = \sqrt{\frac{k_B T_i}{\mu}}, \quad \mu = \frac{M_{em} M_{pert}}{M_{em} + M_{pert}}. \quad (72)$$

In the API, `A_ion` is the emitter mass and optional `A_perturber` defaults to it. The two velocity expressions coincide for equal masses; `fluctuation_rate_model` selects the convention. The dynamic line profile is then given by the Sherman-Morrison form:

$$I(\omega) = \frac{r^2}{\pi} \text{Re} \frac{S(\omega)}{1 - v_i S(\omega)} \quad (73)$$

with the static propagator sum:

$$S(\omega) = \sum_k \frac{p_k}{v_i + \gamma_k + i(\omega - \omega_k)} \quad (74)$$

where $p_k = w_{F,k} w_{\mu,k} |d_k|^2 / r^2$ are the normalized SDT weights and $r^2 = \sum_k w_{F,k} w_{\mu,k} |d_k|^2$. The limit $v_i \rightarrow 0$ recovers the static profile; $v_i \rightarrow \infty$ gives a single Lorentzian (motional narrowing).

By default, `calculate_ffm_profile` uses one shared resonance width $\gamma_k = \gamma(0)$ for all SDTs, matching the historical implementation and the ZEST fast-FFM approximation. With `sdt_frequency_dependent_width=True`, it instead evaluates the upper- plus lower-shell electron width at each post-binning SDT center, $\gamma_k = \gamma_e(\omega_k - E_0) + \gamma_{\text{natural}}$. This optional PPPB-style mode remains $O(N)$ in the sum above. It is distinct from the static solver's pointwise $\gamma_e(E - E_0)$, which varies with observation energy.

SDT binning (`sdt_bin_tol`). The microfield-and-angle quadrature loop typically produces thousands of SDTs per polarization channel, many at nearly the same frequency — the $S(\omega)$ sum above is $O(N)$ per frequency point, so this dominates the FFM runtime. Passing `sdt_bin_tol` (an energy tolerance in eV) merges every group of SDTs within one polarization channel whose frequencies fall in the same bin of that width *before* the Sherman-Morrison solve: intensities are summed and each merged frequency becomes the intensity-weighted mean of its group. In per-SDT-width mode the width is evaluated at that merged frequency. This is an approximation whose error must be checked when `sdt_bin_tol` $\ll (v_i, \gamma_k)$ and reduces N by a factor of 10–100 in practice; `sdt_bin_tol=1e-5` (eV) is a example only, requiring comparison with the unbinned calculation. Leaving it at its default of `None` performs the exact, unbinned calculation.

3.11 Thermal Doppler Broadening

Thermal motion of the radiating ions causes each photon frequency to be Doppler-shifted by $\delta\omega = \omega_0 v_z/c$. Averaging over a Maxwell–Boltzmann velocity distribution yields a Gaussian broadening with $1/e$ half-width

$$\Delta\omega_D = \omega_0 \sqrt{\frac{2k_B T_i}{m_{\text{ion}} c^2}} \iff \Delta E_D = E_0 \sqrt{\frac{2T_i}{m_{\text{ion}} c^2/e}} \quad (75)$$

where T_i is the ion temperature, m_{ion} the emitter mass, and E_0 the transition energy in eV. For H Balmer- α at $T_i = 5$ eV, this gives a $1/e$ half-width $\Delta E_D \approx 0.195$ meV (standard deviation ≈ 0.138 meV, FWHM ≈ 0.325 meV) — several times larger than the normal-Zeeman component shift at 1 T ($\mu_B B \approx 0.058$ meV) and negligible relative to the Stark width at $N_e \sim 10^{23} \text{ m}^{-3}$.

The two calculation paths apply Doppler broadening at different stages.

FFM path (`calculate_ffm_profile`). When `apply_doppler=True` (the default), Doppler broadening is applied *inside* `calculate_ffm_profile` on the energy grid, immediately after the Sherman-Morrison step. The convolution is a zero-padded FFT filter of length $2N$:

$$\tilde{P}_q(k) \longrightarrow \tilde{P}_q(k) \exp\left(-2\pi^2 \sigma_D^2 k^2\right), \quad \sigma_D = E_0 \sqrt{\frac{T_i}{m_{\text{ion}} c^2/e}} = \frac{\Delta E_D}{\sqrt{2}} \quad (76)$$

where k is the discrete-frequency variable (cycles per eV) of the padded rfft and zero-padding to $2N$ avoids circular-wrap artifacts. Set `apply_doppler=False` to obtain the purely Stark-Zeeman FFM profile.

Static profile path (`calculate_static_profile`). When ion temperature is supplied and Doppler is enabled, constant-width components use direct Voigt kernels. Frequency-dependent kernels are Gaussian-convolved on a uniform energy work grid and interpolated back to the requested monotonic coordinates. No width or diagonal-operator request is discarded. Resolve the intrinsic kernels and enlarge the window to check convergence.

Alternatively, leave `Ti_ev` unset during the static calculation and use the standalone wavelength-space helpers afterward:

- `apply_doppler_broadening(wavelengths_nm, profile, Ti_ev, species='H')` — convolves with a Gaussian kernel of width $\Delta\lambda_D$.
- `apply_instrument_broadening(wavelengths_nm, profile, fwhm_nm)` — convolves with a Gaussian instrumental slit function.

Both standalone helpers operate on a *uniform wavelength grid*; compute on or resample to such a grid before calling them. Do not apply the Doppler helper to a profile that already included Doppler inside either solver. The Gaussian kernel is:

$$K_D(\lambda) = \exp\left(-\frac{(\lambda - \lambda_0)^2}{\Delta\lambda_D^2}\right) \quad (77)$$

The complete physical broadening pipeline is therefore:

$$I_{\text{obs}}(\lambda) = \left[I_{\text{SZ}}(\lambda) * G_D(\lambda) \right] * G_{\text{inst}}(\lambda) \quad (78)$$

where I_{SZ} is the Stark-Zeeman profile, G_D is the Doppler Gaussian, and G_{inst} the instrumental Gaussian.

4 Approximations and Numerical Choices

StarkZee is designed to be self-contained and fast enough for exploratory spectroscopy studies. The main approximations are explicit in the solver rather than hidden behind external atomic-structure packages:

1. **Hydrogenic basis and analytic radial functions.** Matrix elements are built from analytic hydrogenic wavefunctions [3],

$$R_{nl}(r) = \sqrt{\left(\frac{2Z}{n}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-Zr/n} \left(\frac{2Zr}{n}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2Zr}{n}\right), \quad (79)$$

giving self-contained r and r^2 matrix elements without requiring Cowan, FAC, or another external structure code. Empirical NIST energies can still be injected for field-free level positions when accurate line centers matter.

2. **Within-shell mixing in production profiles.** The Stark and diamagnetic operators used by the static and FFM solvers retain couplings inside each principal shell but neglect neighboring n manifolds. The experimental `starkzee.multishell` module now provides a standalone globally indexed multi- n basis, signed inter-shell r and r^2 elements, Hermitian Stark and diamagnetic blocks, and coherent target/neighbor/interference oscillator-strength accounting in selected energy windows. Five internal H-beta cases sample 100–1000 T and fixed electric fields from zero through three Holtsmark normal fields at three angles. The minimum tested passing upper basis ranges from the target $n = 4$ shell alone at the low-field, zero-electric-field endpoint to $n = 3 \dots 8$ at 1000 T and three normal fields. These finite-reference tolerances do not transfer automatically to continuous microfield quadrature or other transitions. The module is not yet connected to profile accumulation or external line-profile validation.
3. **Hydrogen-like high- Z energies and dipoles.** Against the PPP manual's Ar XVII example, the retained $O((Z\alpha)^4)$ Hamiltonian reproduces the $2p_{3/2} - 2p_{1/2}$ interval within 2%, but absolute excitation energies are about 0.9–1.0 eV high and the 0.167725 eV $2s_{1/2} - 2p_{1/2}$ Lamb/QED splitting is absent. The four nonrelativistic J-coupled reduced-dipole magnitudes agree within 2%; a consistent rephasing of the $2s_{1/2}$ level reconciles the printed signs. Exact Dirac plus sourced Lamb/QED energies remain pending.
4. **Quasi-static ions with optional dynamics.** The baseline profile treats ionic microfields as static during the radiative event and averages over their distribution. When ion motion is important, the FFM layer modifies the static profile through a fluctuation rate rather than rebuilding the atomic calculation.
5. **Electron-impact broadening.** Fast electrons enter through a Lorentzian impact width. By default (`frequency_dependent_width=True`) the GBK width $\gamma_e(\Delta E)$ is evaluated at the shared observation detuning from the fixed zero-field reference. Setting `frequency_dependent_width=False` fixes the width at the resonance-center value $\gamma_e(0)$. The opt-in `electron_interference=True` mode instead constructs the PPP Appendix-B impact-limit operator, including its upper-lower interference term, and diagonalizes the resulting non-Hermitian coherence generator. It currently

requires $G(0)$ and a finite within-shell closure; neither the closure nor physical accuracy has yet been accepted against PPP/PPPB output.

6. **Microfield orientation quadrature.** The angular integral is evaluated with an N_μ -point Gauss-Legendre rule over $\mu = \cos \theta \in [0, 1]$, using the symmetry of the Hamiltonian under $\mu \rightarrow -\mu$; see Section 3.7.

5 Built-in Reference Models

The `starkzee.models` package provides five comparison lineshape models that share a common call signature. They are grouped into tabulated models (reading precomputed databases) and analytical models. Their curves are diagnostics, not five independent physical benchmarks. `starkzee/models/reference_provenance.json` records which parts are direct source transcriptions and which are StarkZee-specific wrapper operations or approximations.

5.1 Tabulated Models

5.1.1 Stehlé (MMM) — *stehle*

The Stehlé model reads precomputed Stark lineshapes from the Model Microfield Method (MMM) database [18] for hydrogen Lyman, Balmer, and Paschen transitions over a grid of N_e and T_e values. The tables contain the *unmagnetized* ($B = 0$) Stark profile **without fine structure** [18], as a function of a reduced detuning $\Delta\omega/F_0$ (Holtsmark normal field units). The pipeline is:

1. Linear interpolation over the required (N_e, T_e) table corners. Exact coordinate nodes use only that source node. Positive detunings use log-log interpolation by default; `interpolation='finterp'` selects the distributed Fortran-style three-point rule.
2. Thermal Doppler broadening via FFT convolution.
3. Normal Zeeman splitting applied *after* the table: the whole Stark+Doppler profile is rigidly copied to $\omega \pm \mu_B B/\hbar$ (the normal-Zeeman component shift – half the electron Larmor/cyclotron frequency $\omega_L = eB/m_e$ used elsewhere for the plasma dynamical cutoffs, not ω_L itself) with angle-dependent π/σ weights:

$$I(\theta) = \sin^2\theta I_\pi + \frac{1+\cos^2\theta}{2} (I_{\sigma+} + I_{\sigma-}) \quad (80)$$

The Stark and Zeeman effects are therefore treated as **separable**. This is valid only when the Zeeman splitting $\mu_B B$ is much smaller than the Stark width. The bundled file contains exactly 84 transitions: $n_l = 1, 2, 3$, with $n_u = n_l + 1, \dots, 30$. Requests outside that set are rejected. Its ten temperature nodes span 2500–1,259,600 K; density endpoints and cell availability depend on transition and temperature. Exact endpoints are accepted when their source cell is available, while extrapolation beyond stored coordinates is rejected. H, D, and T select the wrapper line center and Doppler mass but share these hydrogen Stark tables; this is not an isotope-specific MMM calculation. Normal-Zeeman post-processing accepts nonnegative B , but does not make the field-free database a coupled Stark–Zeeman model.

The bundled NetCDF file has no embedded global or variable provenance/unit attributes. It is nevertheless byte-identical to the NetCDF at pinned `pystark` commit 9a8782a,

whose tree contains the CDS/VizieR VI/98A raw distribution and conversion script. Checksums, units, paths and unresolved license and uncertainty fields are recorded in `starkzee/data/reference_tables.json`. This traceability and the implementation-level inventory checks do not replace comparison with held-out physical data.

5.1.2 Rosato — *rosato*

The Rosato database [19] solves the Stark-Zeeman problem *jointly* and tabulates the result with B as an explicit table axis ($B \in \{0, 1, 2, 2.5, 3, 5\}$ T). The pipeline is:

1. Interpolation over the (N_e, T_e, B) grid. The B -interpolation is *scaled*: before blending two bracketing profiles at B_0 and B_1 , each profile's detuning axis is stretched by B_{node}/B . This exploits the fact that Zeeman splitting scales linearly with B , aligning the σ components before interpolating.
2. Angle dependence from *two real tables* (parallel and perpendicular to $\vec{B}$), blended as $I = I_{\parallel} \cos^2 \theta + I_{\perp} \sin^2 \theta$. Both tables carry the correct π/σ lineshape and width — the angle enters as a physical interpolation, not just an amplitude weight.
3. Thermal Doppler broadening via FFT convolution.

Rosato supports D Balmer lines with upper $n = 3 \dots 7$, and an approximate H mapping. Density covers $1\text{e}13..1\text{e}16 \text{ cm}^{-3}$, effective table temperature $0.316..31.6 \text{ eV}$ and $B=0..5 \text{ T}$, including endpoints. The effective temperature is T_i for D and $2*T_i$ for H; T_e is not a table coordinate. The corrected code uses four-corner logarithmic density/temperature interpolation, retains Doppler tails and normalizes output per wavelength metre. No interpolation accuracy outside the tables is claimed.

Rosato source nodes can contain negative intensities. By default, `negative_table_policy='warn'` warns and preserves signed values through interpolation, Doppler smoothing and normalization. Explicit `'raise'` rejects affected cells; `'clip'` projects values to zero with a warning. Clipping can bias noisy data and is not validated table repair. Reference wrappers return density per wavelength metre on air-centered axes; main-solver profiles are per eV. Convert coordinates and spectral densities and match the finite normalization window before comparing amplitudes.

The Rosato NetCDF has no embedded attributes, but an exhaustive provenance check establishes that its 3,000,000 detunings and 3,000,000 intensities equal the 3,000 raw profiles at pinned `pystark commit 9a8782a` after numeric parsing. The archived reader declares density in cm^{-3} , temperature and detuning in eV, and magnetic field in tesla. Intensity is interpreted as eV^{-1} from the area-normalized line-shape convention, rather than from an explicit source declaration. The original database-transfer URL/date, license, numerical uncertainty, and intended treatment of signed fluctuations remain unknown. The full record and optional upstream verifier are in `starkzee/data/reference_tables.json` and `scripts/verify_reference_table_sources.py`.

5.2 Analytical Models

5.2.1 Lomanowski-width sensitivity model — *lomanowski*

This StarkZee-specific sensitivity model uses the power-law Stark FWHM from Lomanowski *et al.* Eq. (1) and Table 1 [20]:

$$\Delta\lambda_S = c_{n_u n_l} N_e^{a_{n_u n_l}} T_e^{-b_{n_u n_l}} \quad [\text{nm}] \quad (81)$$

Coefficients (a, b, c) are tabulated for H/D Balmer and Paschen lines up to $n_u = 9$. The subsequent common-width modified-Lorentzian/Gaussian pseudo-mixture is local to StarkZee: it is not the profile prescribed by the cited paper and is not a `pystark` routine. Its mixing-polynomial provenance has not been recovered. The wrapper adds separable normal-Zeeman splitting, so the function does accept B ; it remains a sensitivity approximation rather than an independent benchmark.

5.2.2 Parameterized Stehlé — `stehle_param`

The direct Lomanowski parameterization of the Stehlé/MMM tables [20]: Eq. (1) and Table 1 give the Stark FWHM and coefficients, while Eq. (2) gives the modified-Lorentzian profile. StarkZee then performs frequency-domain Doppler convolution and applies a separable normal-Zeeman triplet. The paper reports its fit domain as $10^{19} \leq N_e \leq 10^{21} \text{ m}^{-3}$ and $1 \leq T_e \leq 10 \text{ eV}$, with under-10% FWHM error against the original tabulations within that domain; this is not a guarantee for the complete profile or for extrapolation.

5.2.3 Voigt — `voigt`

A Faddeeva-evaluated Voigt profile using the Griem α_{12} Stark half-width [7] combined with thermal Doppler broadening. Serves as a quick estimate when only order-of-magnitude Stark width accuracy is required.

5.3 Comparison with StarkZee

Feature	StarkZee	Stehlé	Rosato
Magnetic field	Full simultaneous diagonalization of $H = H_A + V_E$	$B = 0$ in tables; added as rigid triplet after	B is a table axis
Fine structure	Yes — spin-orbit, MV, Darwin	No (degenerate hydrogenic)	In tables
Electron broadening	GBK semi-classical, frequency-dependent	Unified theory (more rigorous far wings)	In tables
Ion dynamics	Quasi-static + optional FFM	Model Microfield Method (MMM) – distinct from StarkZee’s FFM, not “static plus some dynamics”	In tables
Microfield	Potekhin (default)	Holtmark/Hooper variant	Holtmark variant
B -treatment	Intrinsic to Hamiltonian	External convolution	B -scaled interpolation

The most important physical distinction at $B \neq 0$: StarkZee diagonalizes $H = H_A + V_E$ *simultaneously* for all field strengths. When $\mu_B B \sim 3n e a_0 F$ (Zeeman and Stark splitting comparable) the eigenstates are genuine Stark-Zeeman hybrids — neither pure Zeeman nor pure Stark states. No post-processing convolution can reproduce this mixing. Separable Zeeman post-processing cannot reproduce this mixing. Rosato tables already contain joint Stark-Zeeman physics; their interpolation error requires a separate convergence or held-out-node comparison.

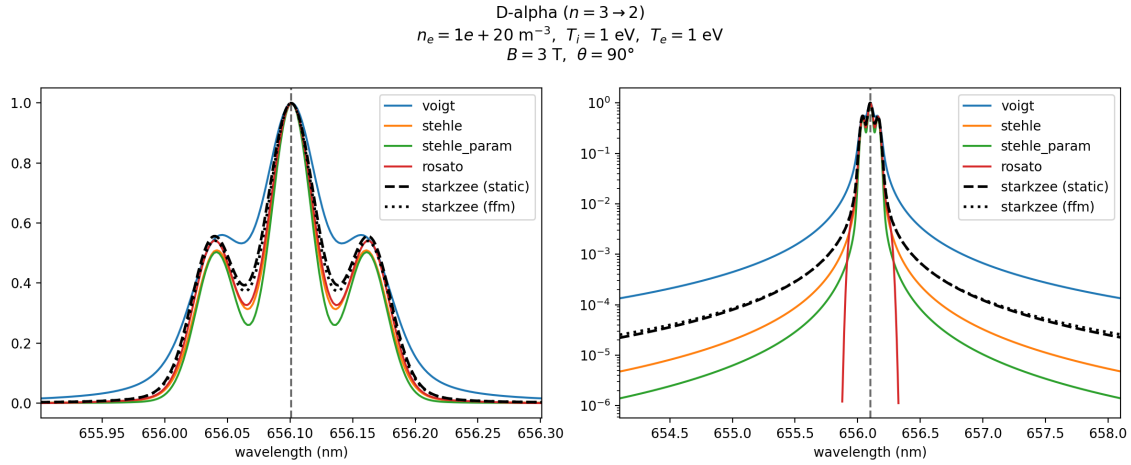


Figure 1: Current D-alpha model comparison generated from this working tree. The analytical and Stehle wrappers add a separable normal-Zeeman triplet; Rosato tables contain joint Stark-Zeeman physics. Rosato values below 10^{-6} of its peak are omitted from this visualization only; the model output and source tables are unmodified. This is a model comparison, not an independent accuracy benchmark.

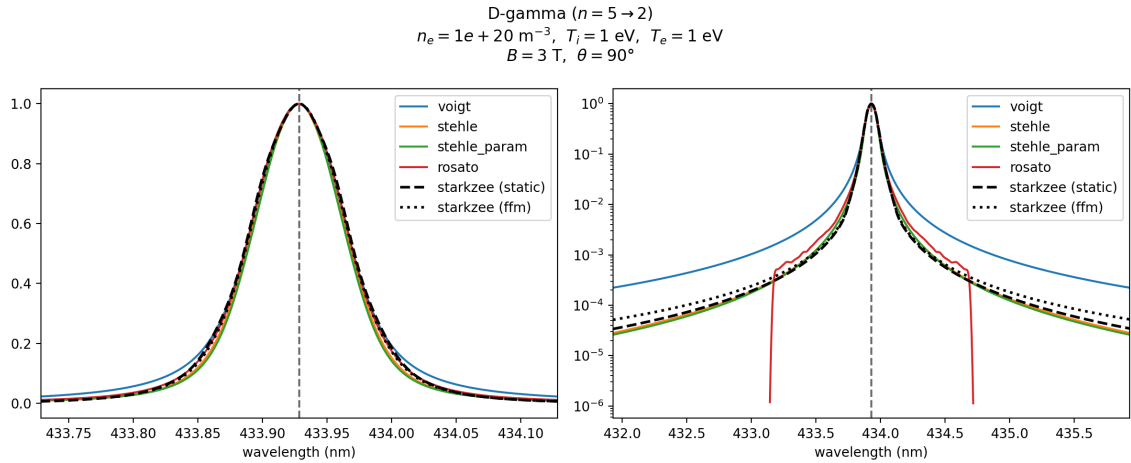


Figure 2: The corresponding D-gamma ($n = 5 \rightarrow 2$) comparison at the same plasma conditions and with the same display-only Rosato cutoff.

A Physical Approximations and Domains of Validity

Approximation	Mathematical form	Domain of validity	Failure mode
Within-Shell Isolation ($\Delta n = 0$)	$V_E = \text{diag}_n(V_E)$, no $n \rightarrow n \pm 1$ coupling	Stark shift $\ll$ shell spacing $2Z^2\text{Ry}/n^3$	Inter-shell mixing or merging; threshold depends on state, charge and field statistics.
Quasi-Static Ion Microfields	$\vec{F}_{\text{ion}} = \text{const}$	Fluctuation energy $\hbar\nu_i \ll \Delta E_S$ (convert to eV)	Ion dynamics / motional narrowing; use FFM.
Semi-Classical Electron Impact	Lorentzian with GBK width γ_e	$\lambda_{dB} \ll$ impact parameter ρ	Quantum close-coupling needed for cold dense plasmas.
Dipole Approximation	$V_{\text{rad}} \propto \vec{r} \cdot \vec{E}$	$\lambda \gg \langle r \rangle_n \sim n^2 a_0 / Z$	Quadrupole/octupole for high- n Rydberg states.

Table 1: Key physical approximations, their domains of validity, and failure modes.

B Physics Explanation of Use Cases and Operational Limits

B.1 Magnetic-field and density limits

The normal-Zeeman component shift is 0.05788 meV per tesla. Its importance relative to fine structure and Stark mixing depends on transition, charge and microfield distribution, not a universal field interval. The production quadratic-Zeeman and Stark operators are retained within a single shell only. The experimental multi-shell reference matrices are not yet used by either profile solver; higher relativistic terms are also absent. Diagonalizing either finite matrix does not validate compact-object spectra.

At fixed temperature and charge, lowering density collapses the physical field scale toward zero, but the reduced-field distribution need not become a delta. Doppler dominance also depends on ion temperature and radiative lifetime. At high density compare Stark and diamagnetic couplings with shell gaps and atomic extent with perturber spacing. Independently compare fluctuation energies with component widths: high density alone does not imply quasi-static breakdown. Check matched static/FFM calculations and converge grid spacing, window, field cutoff, field resolution, angular quadrature and binning independently.

C Bibliography Audit and Historical Snapshot

The machine-readable file `starkzee/data/bibliography.json` records canonical metadata and a verification URL for all 22 works represented by the 21 citation keys below. Titles, author lists, publication containers, years, volumes/issues, page ranges or article numbers, and DOI identifiers were audited. The NIST entry is release 5.12 (2024), which was current when the archived D I and T I queries were downloaded on 13 July 2026. The H I table

lacks an archived query URL and acquisition timestamp, so its exact NIST release remains unknown.

The standalone manual audited on 7 September 2026 is tied to immutable StarkZee commit 24118d1fdee16d3c5aef98992809c5a0db2faea0. Its historical docs/manual.tex has SHA-256 dbfcb4766218bbb1d458982865bc22d6b5158b99e1359135908d4bf8f0ef89b6, exactly the digest recorded in the original audit. That commit also contains the compiled manual, referenced comparison figure, figure generator, model implementation, and all three runtime datasets. Exact blob IDs, sizes, hashes, and immutable URLs are recorded in starkzee/data/historical_provenance.json; a full clone can verify them with python scripts/verify_historical_provenance.py. This establishes repository ancestry, not physical accuracy or external-data licensing.

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